

PROTOTYPE PHASE 01 / SAFETY WAYFINDING

EvacWise

A dynamic route planner for modeled venue evacuations.

SIMULATION NOTICE

This prototype uses fictional venue data and is not certified for real emergency use. Do not rely on it for actual evacuations.

From fixed plans to adaptable routes

The Static Limit

Traditional evacuation maps are fixed anchors. They teach one path, assuming all exits are clear and every corridor is accessible.

- + Assumes perfect visibility and path availability
- + Requires manual mental rotation in high-stress states
- + Static info becomes dangerously obsolete during incidents

The EvacWise Shift

We treat the venue as a dynamic graph. When conditions change—a door is locked or a hallway is blocked—the route recalculates.



Visual Anchor: Familiar wayfinding principles meet dynamic data. Source: U.S. Fire Administration / FEMA.

Four steps to a route

01 / ORIENT

Select Starting Location

Pick a room or junction on the interactive map to establish the origin node.

02 / CONSTRAIN

Set Conditions

Apply modeled obstructions: block corridors, close exits, or toggle step-free access.

03 / COMPUTE

Route Recalculation

FastAPI runs Dijkstra over the venue graph, excluding unavailable edges instantly.

04 / GUIDE

Present Directions

Review the exit, distance, walk time, turn-by-turn steps, and alternate paths.

A stateless, immediate feedback loop that turns simulation into actionable guidance.

Test what changes

SCENARIO A / EVACUEE

Clear Path Navigation

The baseline state. Recalculate routes as you move through the venue, ensuring you're always heading toward the most efficient exit.

MODE: EVACUEE

SCENARIO B / ACCESS

Step-Free Requirement

Simulation of limited mobility. Dijkstra's weights are adjusted to strictly exclude stair nodes, prioritizing ramps and level corridors.

VAR: AVOID_STAIRS=TRUE

SCENARIO C / INCIDENT

Path Obstructed

Manually block any modeled corridor. The engine identifies the next best reachable exit instantly, preventing entry into dead zones.

VAR: BLOCKED_EDGES=[...]

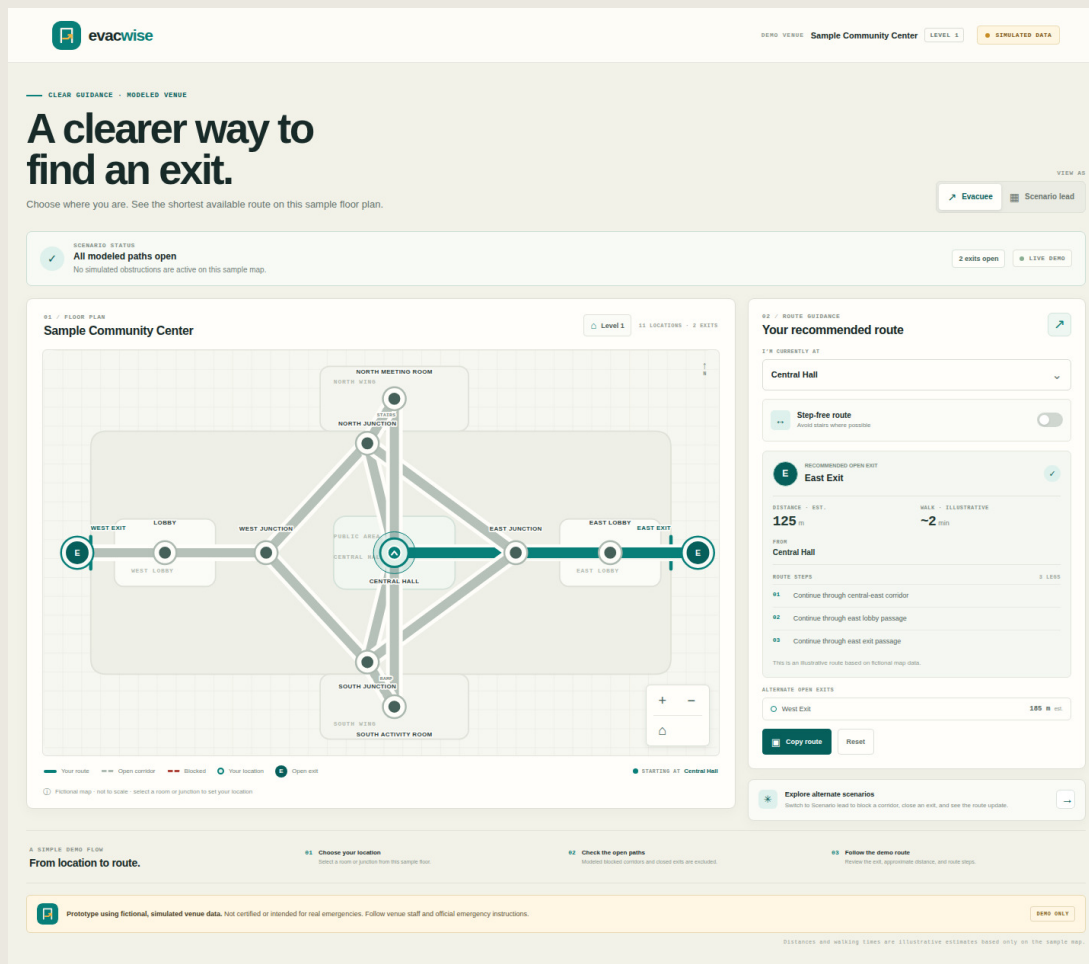
SCENARIO D / CLOSURE

Exit Compromised

Close any modeled exit. The system validates if any other route exists and provides an explicit "No Path" state if all exits are unreachable.

VAR: CLOSED_EXITS=[...]

A map-first prototype



UI / INTERACTIVE SVG

Wayfinding centerpiece

A labeled floor plan with room shapes, corridor types, and route arrows that update in real-time.

UX / DUAL VIEWS

Evacuee & Coordinator

Switch between the simple evacuee guidance and the lead view for scenario testing.

ACCESSIBILITY

Inclusive design

Keyboard-usable nodes, screen-reader support, and redundant visual cues for status.

Small stack, clear responsibilities

Technical Stack

FRONTEND	Vanilla JS / SVG
SERVICE	FastAPI / Uvicorn
ROUTING	Dijkstra's (Stateless)
STORAGE	None (Graph in code)
PACKAGE	Docker (Linux Slim)

Primary API Interface

ENDPOINT / POST /API/V1/ROUTES

```
{
  "venueId": "demo-building",
  "startNodeId": "central",
  "blockedEdgeIds": ["n-c"],
  "closedExitIds": [],
  "avoidStairs": false
}
```

Stateless requests ensure every route matches current simulation parameters exactly, with no stale data.

What it proves—and what it doesn't

Current Prototype

- Fictional static venue graph
- Manual scenario simulation
- Illustrative distances & times
- Stateless Dijkstra routing

Required for Deployment

- Live building sensor integration
- Official incident feed ingest
- GPS / IPS indoor positioning
- Emergency service certification

Primary Objective

- Proving recalculation efficiency
- Validating wayfinding UI/UX
- Testing accessibility logic
- Defining the API contract

SAFETY BOUNDARY

EvacWise is a simulation tool. It does not send alerts or represent live building conditions. In emergencies, prioritize official instructions and venue staff above all digital guidance.